

# Ben Bubnick

DATA ENGINEER · DATA SCIENTIST · ANALYTICS SPECIALIST

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## Summary

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Experienced Data Engineer with a strong track record of delivering scalable health data pipelines across cloud platforms, automating validation processes, and documenting workflows.

## Experience

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### Health Data Movers

*AWS: S3, Snowflake, SQLMesh, Python, SQL, Docker*

#### DATA CONSULTANT

*Mar. 2025 - Present*

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Developed SQLMesh models for ingesting and transforming payer and RCM data in Snowflake.
- Designed and built reusable ETL pipelines to support revenue cycle integrations, reducing onboarding time.
- Mapped complex healthcare data schemas from Practice Suite into Adonis' data model.
- Created testable, modular SQLMesh models, reducing regression errors and accelerating iteration cycles.
- Standardized and documented transformation logic across clients in order to cut onboarding documentation.
- Established validation and audit checkpoints (e.g., row counts, null checks, referential integrity).
- Translated business requirements into technical specifications to improve delivery predictability.
- Built detailed integration playbooks to be used as the internal standard for onboarding new vendors and partners.

### Arkos Health

*AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter*

#### SENIOR DATA ENGINEER

*Jun. 2023 - Jan. 2025*

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Designed and implemented an AWS data lake, improving data retrieval times by 50%.
- Automated CI/CD, reducing deployment times by 50% and increasing success rates by 15% over the past year.
- Created metadata reports, reducing troubleshooting time by over 90% and enhancing workflow transparency.
- Integrated Jupyter into reporting, boosting report interactivity and increasing stakeholder satisfaction by 30%.
- Created comprehensive data lake documentation, reducing onboarding time for new team members by 40%.
- Implemented Jira and GitHub, enhancing task tracking and team coordination by 25% within three months.
- Enhanced cross-departmental collaboration, increasing data-driven decision-making by 15% in six months.
- Mentored junior team members, improving their performance metrics by 30% in six months.
- Led goal-setting sessions, improving goal alignment and tracking accuracy by 30% in the first quarter.
- Initiated stakeholder feedback sessions, improving deliverable relevance and accuracy by 25% in three months.

### Merative

*Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps*

#### SENIOR DATA SCIENTIST

*Jul. 2022 - Mar. 2023*

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification initiative across 200+ million patient records.
- Designed and implemented an automated testing framework for ETL pipelines for validation coverage.
- Validated and optimized Azure-based data workflows, ensuring continuity and fidelity during the transition from legacy HDFS systems to cloud architecture.
- Conducted end-to-end system testing for Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations.
- Collaborated with engineering teams across disciplines to refine architectural designs.
- Documented test plans and platform migration procedures for improving onboarding for future QA engineers.
- Built regression test suites and data comparison tools, detecting schema drift and transformation errors early in the development lifecycle.
- Ensured compliance with data governance and security protocols to reduce audit risks and support HITRUST readiness.

**ADVISORY DATA SCIENTIST**

Nov. 2021 - Jul. 2022

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Created AI Surgical Scheduling Optimization tool currently evolving into new offering.
- Led delivery on tailored analytics and reporting project for client administrative data.
- Developed data migration validation analysis on 150+ billion patient records.
- Developed surgical smart case card using combined CNN/RNN model.

**SENIOR DATA CURATION ARCHITECT**

Dec. 2020 - Nov. 2021

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Developed clinical data integration of 100+ billion structured & semi-structured records.
- Developed topic modelling project on nearly 6 million semi-structured provider notes data.
- Curated 25 pages of workflow documentation and mentored new hires in AI tuning process.
- Created COVID-19 hospitalization prediction tool with Machine Learning algorithms.

**DATA CURATION ARCHITECT**

Nov. 2017 - Dec. 2020

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.
- Reduced mapping rework by 45% with through new knowledge-driven initiatives.
- Led mentoring program to increase 4,500 billable hours over two year period.

**SOFTWARE DEVELOPER**

Sep. 2016 - Nov. 2017

Developed internal resources that reduced costs and improved onboarding efficiency.

- Reduced time spent on knowledge transfer by 80% by rewriting project documentation.
- Eliminated need for a \$30,000 annual contract by developing data curation tool.
- Reduced cost-per-hire by \$5,210 by re-developing education program.

**DATA SCIENTIST**

Nov. 2015 - Sep. 2016

Led integration efforts and process automation initiatives.

- Managed team of 12 remote data science contractors for 36 integration projects.
- Created over 125 new knowledge base articles, making up 30% of the team's online content.
- Developed automated data integration tool to cut out 28% unnecessary development time.

**Explorys**

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

**DATA SCIENTIST**

Jun. 2015 - Nov. 2015

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Developed cron-based widget to define blackout periods for client data pulls.
- Performed troubleshooting on production issues across multiple environments.
- Principal integration specialist for 10+ client database warehouses.

**Multiple Institutions****ADJUNCT PROFESSOR**

Aug. 2010 - May. 2019

Taught physics & astronomy lecture series and lab courses.

- Consistently exceeded student evaluation benchmarks, surpassing institutional feedback goals by an average of 16% in 2017 and 17% in 2018, reflecting strong teaching effectiveness and student satisfaction.
- Improved student comprehension and course navigation by reorganizing online instructional materials, resulting in a 28% reduction in reported confusion and fewer clarification requests during office hours.
- Developed and published over 700 hours of recorded lectures and lab content, enhancing accessibility and supporting flexible, on-demand learning for both traditional and non-traditional students.
- Implemented Just-in-Time Teaching (JiTT) strategies for underperforming students, leading to an average grade improvement of 18% by the end of term.
- Designed interactive lab lectures based on the Interactive Lecture Demonstration (ILD) method, increasing student engagement and concept retention in physics instruction.
- Promoted collaborative learning through Peer Instruction and ILD techniques, effectively addressing real-time student misconceptions during physics lab sessions.

## AmTrust Financial Services

VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

### SOFTWARE ENGINEER

Mar. 2014 - Jun. 2015

Developed MVC components and web forms for insurance policy software.

- Led development on core insurance policy software, serving as principal developer for a new product offering and contributing to two major architectural initiatives in 2015.
- Analyzed and optimized performance across 65+ ISO rating-based web forms using R, identifying bottlenecks and reducing page load times.
- Championed and implemented a unit testing framework for multi-tier MVC components, improving code quality and reducing deployment errors across the dev team.

## PPG Industries

VB.NET, C#, SQL, C++, FORTRAN

### COLOR SCIENTIST

Apr. 2012 - Jun. 2013

Developed MVC components and web forms for insurance policy software.

- Improved color matching classification algorithm accuracy by 40% by developing and applying advanced numerical methods, enhancing product performance across automotive coatings.
- Developed cross-instrument correlation software to support a risk analysis initiative, enabling consistent color measurement across varying hardware platforms and improving QA reliability.
- Supported the creation of an industry-leading automated color matching facility, developing numerical tools and methods to improve formulation quality and pigment data analysis in high-variance OEM color environments.

## Awards

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|             |                                                                                          |                |
|-------------|------------------------------------------------------------------------------------------|----------------|
| 2018 - 2022 | <b>Gold Champion</b> , For 1,477 classroom hours and earning 43 digital credentials      | IBM            |
| 2018        | <b>Lab Services Award</b> , For coordinating internal teams to deliver solutions on time | IBM            |
| 2017        | <b>Eminence and Excellence Award</b> , For developing a new data curation tool           | IBM            |
| 2012        | <b>Color Lab Automation Award</b> , For expanding data collection and analysis reports   | PPG Industries |
| 2010        | <b>Mary J. Hanna Fellowship</b> , For STEM students pursuing research-based Masters      | UC             |

## Volunteer

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### CSforAll

#### INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

### HIT in the CLE

#### COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

### Code for America

#### ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

### T.C.P. World Academy

#### TUTOR

Sep. 2008 - May. 2009

Our goal was to teach students how to understand the value of their learning experience and becoming academically involved by participating in society simulation activities for higher learning.

## Talks

### AI Hackathon 2020: Surgery Scheduling Optimization

*"An algorithm uses past utilization and current waitlist info to predict surgery duration.*

Dec. 2020

IBM

### REACH Initiative: IBM Watson Care Insights

*Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.*

2019

IBM

### Content & Data Analytics Learning Exchange

*Introduction of implementation of machine learning algorithms using real-world health data analytics.*

2018

IBM

### 2018 Innovation Research Interchange Fall Networks Conference

*Using NLP to extract insight from unstructured data and transform it from a data lake.*

Sep. 2018

IRI

### Cross Team Training Series

*A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.*

2016

EXPLORYS

## Publications

### Surgical Scheduling Optimization and Procedure Duration Prediction <sup>a</sup>

*Improvement on traditional fixed ratio methods for total procedure time estimation.*

Dec. 2020

IBM

### Process and Requirements for CCD Data Integration

*Processes and requirements for overcoming the many difficulties with CCD data curation.*

Oct. 2017

IBM

### AI Paint Formula Prediction from Spectroscopic Measurements <sup>b</sup>

*Naive bayesian color and effects matching algorithm of complex automotive paint formula.*

May. 2015

SHINYAPPS.IO

### Characterization of Surface Coatings using RT Models for Color Matching

*Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.*

Feb. 2013

USPTO

### Massive Stellar Clusters in the Disk of the Milky Way Galaxy <sup>c</sup>

*Spectroscopic and Photometric analysis of two open clusters in the galactic plane.*

Dec. 2010

OHIOLINK ETD

### VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

*Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.*

May. 2008

VIZIER

### Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107

*New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).*

Feb. 2008

PASP

### A Novel Method for Low-Toxic Photo-Etch Printing <sup>d</sup>

*Application of aquatint etching techniques to photo-etch procedures.*

May. 2003

MAPC

## Education

2022 **PGCert.**, Data Science Professional<sup>a</sup>

IBM

2015 **PGCert.**, Data Science<sup>b</sup>

Johns Hopkins University

2010 **Master of Science**, Astrophysics<sup>c</sup>

University of Cincinnati

2004 **Bachelor of Fine Arts**, Printmaking<sup>d</sup>

Ohio University